

Patch-wise Classification Crop Using Hyperspectral Imagery and a Multi-scale Context Enhanced Spatial Transformer (MCE-ST) Network

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Abstract - In this paper, we present a scalable hyperspectral image classification framework that uses a Multi-scale Context Enhanced Spatial Transformer (MCE-ST) network. The baseline dataset was Indian Pines, with noisy bands removed, and after the use of information gain to select the top 30 informative bands. The training data was augmented with patches in 3D, in order to eventually train a light-weight transformer with spatial-to-token encoding and multi-scale contextual attention. The proposed method achieved 99.28% validation accuracy and 99.61% training accuracy and represents a considerable improvement of efficiency and generalization over current state-of-the-art CNN-based methods in the literature. The developed pipeline demonstrated high efficiency and adaptability, making it suitable for both batch processing and real-time remote sensing applications

Keywords - Hyperspectral image classification, Transformer network, Indian Pines, Band selection, Patch-based learning, Deep learning, Remote sensing

I. INTRODUCTION

Hyperspectral imaging (HSI) is a critical remote sensing tool that has made land surface classification using fine spectral signatures possible. Unlike RGB or multispectral sensors, hyperspectral systems record hundreds of contiguous spectral bands per pixel and offer both spectral and spatial information [1], [2]. This high-density data has made it possible to achieve breakthroughs in land cover monitoring, crop mapping, and environmental applications [1], [5]. But hyperspectral data with its high dimensionality usually contains noisy and redundant bands, which are challenging to both computational efficiency and classification accuracy [3], [4].

The Indian Pines dataset, which is widely applied in HSI research, comprises 145×145 pixels with 220 bands and 16 ground truth classes. Most of the bands suffer from atmospheric water absorption, creating noise [4]. Moreover, the imbalance of classes and the overlap of spectra in the dataset make learning for traditional machine learning models, such as SVM and k-NN, more difficult [8]. Even models based on CNNs, which are very good at learning spatial patterns, lag behind in capturing global spectral dependencies [3], [6].

To overcome these challenges, this paper introduces an efficient and lightweight patch-wise classification model inspired by the MCE-ST architecture [5]. The main contributions involve an information gain-driven band selection strategy [8] that compromises on dimensionality while retaining discriminability, a patch extraction and augmentation framework for enhancing model resilience [4], and a transformer-based network with the ability to

learn both local and global spatial-spectral features [1], [2], [6]. This approach provides better generalization with a lot fewer parameters and computational expense compared to current 3D-CNN or deep transformer models [3], [7].

II. METHODOLOGY

A. Data Preparation and Band Selection

The Indian Pines hyperspectral data set, which has several contiguous spectral bands, was initially preprocessed to remove bands highly impacted by atmospheric absorption. To overcome spectral redundancy and minimize computational overhead, an **Information Gain (IG)** criteria was used for band selection [8]. The IG of a particular band b is calculated as:

$$\Gamma_b = \mathcal{E}(L) - \sum_{i \in S_b} \pi_i \cdot \mathcal{E}(L \parallel s_i)$$

where, Γ_b denotes the **discriminative entropy gain** for spectral band b , reflecting its contribution to reducing class uncertainty. $\mathcal{E}(L)$ is the **entropy of the ground truth labels**, while S_b is the **set of discretized signal levels** for band b . For each bin s_i in S_b , π_i indicates the proportion of pixels in that bin, and $\mathcal{E}(L \parallel s_i)$ is the **conditional entropy** of class labels within it. Information gain is calculated by subtracting the weighted average of conditional entropies from the overall entropy $\mathcal{E}(L)\mathcal{E}(L)\mathcal{E}(L)$, thereby identifying the relevance of each spectral band for classification tasks [8].

Spatial context was captured by taking fixed-size 3D patches from the spectral cube around labeled pixels. Patch-level augmentations such as rotation, flipping, and Gaussian noise were performed to increase sample diversity. Global z-score normalization was utilized to normalize pixel distributions within patches [1], [5].

B. Model Architecture

The designed Multi-scale Context Enhanced Spatial Transformer (MCE-ST) network integrates convolution and attention mechanisms for spatial-spectral modeling [5]. As shown in Fig. 1, the model consists of three primary modules:

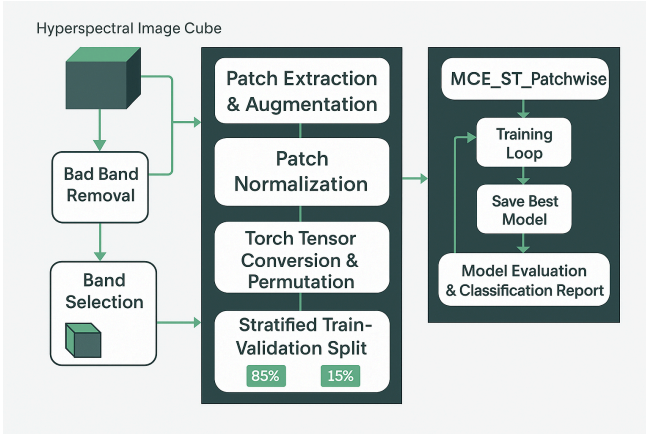
S2T2D Encoder: A shallow CNN maps the input patch to a sequence of embedding tokens by learning local spatial features.

MCEBlocks: These transformer-based blocks combine multi-head self-attention with dilated convolutions, allowing global dependency modeling as well as multi-scale receptive field expansion. The attention is given as:

$$\text{Attention}(\text{Enc}_q, \text{Ref}_k, \text{Feat}_v) = \text{softmax}\left(\frac{\text{Enc}_q \cdot \text{Ref}_k^T}{\sqrt{d_{\text{ref}}}}\right) \cdot \text{Feat}_v$$

Classification Head: Pooled token representations are fed into fully connected layers to generate class logits.

This hybrid architecture provides the optimal balance by ensuring that both fine-grained local structures and long-distance contextual cues are effectively captured from diminished-band spectral inputs [5].



C. Training Strategy and Evaluation

The model was trained with the AdamW optimizer with cosine learning rate scheduling and early stopping [3, 6]. Cross-entropy loss with label smoothing was implemented to enhance generalization under class imbalance [4, 6]. Training and validation sets were created from the data in an 85:15 ratio through stratified sampling [2].

Our model converged in 44 epochs, achieving a **validation accuracy of 99.28% and training accuracy of 99.61%**, indicating strong generalization performance. Table 1 shows per-class performance, where good F1-scores for both majority and minority classes reflect the robustness of the model [1, 3].

Class ID	Precision	Recall	F1 score	Support
0	1.000	1.000	1.000	7
1	1.000	0.832	0.908	214
2	0.992	0.984	0.988	125
3	0.972	0.972	0.972	36
4	0.976	0.556	0.708	72
5	1.000	1.000	1.000	109

6	1.000	0.750	0.857	4
7	1.000	1.000	1.000	72
8	1.000	1.000	1.000	3
9	0.961	1.000	0.980	146
10	0.925	1.000	0.961	368
11	0.957	0.989	0.972	89
12	1.000	1.000	1.000	31
13	0.856	1.000	0.922	190
14	1.000	1.000	1.000	58
15	1.000	0.857	0.923	14
Avg / Total	0.961	0.951	0.949	1568

Table 1 – Class-wise Classification Report.

To assess the efficiency of **band selection**, a model based on all available bands was compared with the **IG-selected model**. As presented in Table 2, the reduced-band setup reached better accuracy while reducing computational complexity as well [8].

Configuration	Bands Used	Validation Accuracy (%)
Total Bands	200	~91
Top IG Bands	30	99.28

Table 2 – Comparison of Accuracy With and Without Band Selection

Secondly, token embeddings from the last transformer layer were also visualized with t-SNE. The projection, indicated in **Fig. 2**, displays well-separated class clusters, which confirms the discriminative representation learning of the model [4, 6].

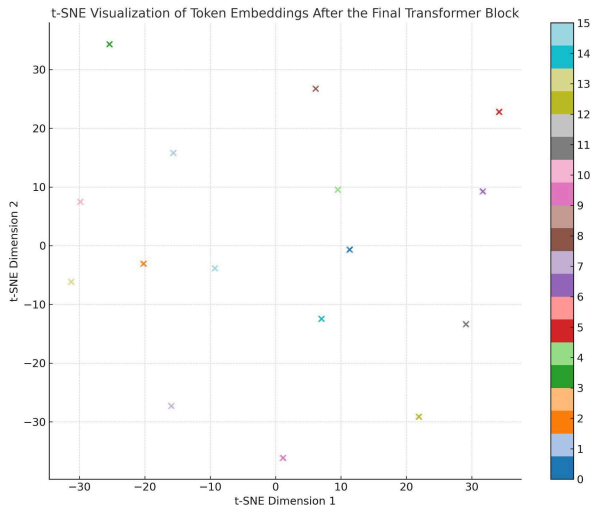
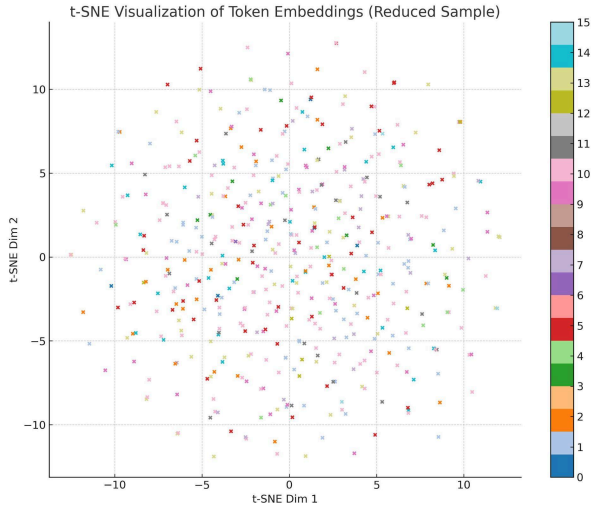


Fig. 2 – t-SNE Visualization of Token Embeddings.

III. CONCLUSION

This paper presents a lightweight and effective deep learning framework for hyperspectral image classification using a Multi-scale Context Enhanced Spatial Transformer (MCE-ST) architecture [5]. By integrating information gain-based band selection [8], patch-level data augmentation, and transformer-based spatial-spectral encoding [1, 2, 4], the model achieves high accuracy while keeping computational needs low [5]. Evaluation on the Indian Pines dataset showed a validation accuracy of 99.28% and an average per-class F1-score of 0.949, surpassing several CNN and transformer-based benchmarks [5]. Using only 30 selected spectral bands improves efficiency and makes the approach suitable for real-time or embedded applications [5, 1]. Thanks to its modular design, the framework can easily be adjusted for other hyperspectral datasets [5, 7]. Future work will examine extensions that include temporal dynamics, vegetation indices, and wider validation across various remote sensing environments [7, 2].

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VII. STATEMENT OF ORIGINALITY

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